SYSTEMS UNDER PRESSURE — HONORS RUBRIC

Grade 11 Honors Algebra 2 Mathematical Modeling Investigation

INDIVIDUAL RUBRIC (6 STANDARDS)

STANDARD	4 – EXCEEDS EXPECTATIONS	3 – MEETS EXPECTATIONS	2 – APPROACHING EXPECTATIONS	1 – BELOW EXPECTATIONS
1 SEQUENCES & SERIES MODELING	• Constructs and uses recursive, explicit, arithmetic, geometric, or series-based models with sophisitcated understanding. • Performs accurate calculations and explains repeated or cumulative change clearly in context. • Uses equations, tables, graphs, and calculations to make and defend meaningful predictions. • Analyzes long-term behavior, sums, convergence/divergence, and identifies where the model breaks down or becomes unrealistic.	• Constructs and uses sequences or series appropriately. • Performs calculations accurately most of the time. • Explains repeated or cumulative change reasonably in context. • Uses mathematical models to support predictions or conclusions.	• Uses sequences or series with limited explanation or inconsistent calculations. • Predictions or interpretations may be unclear or partially developed. • Connections to the investigation may be weak.	• Little or no meaningful use of sequences or series. • Calculations, equations, or reasoning are incomplete or inaccurate. • Work is disconnected from the investigation.
2 EXPONENTIALS & LOGARITHMS	• Constructs exponential or logarithmic equations/models appropriately and justifies model choice. • Calculates and interprets ratios, rates, thresholds, half-life, logs, or critical values accurately. • Uses equations, graphs, and calculations to make predictions and analyze sensitivity to initial conditions or parameters. • Critiques exponential assumptions and identifies limitations or points of failure.	• Uses exponential or logarithmic reasoning appropriately. • Performs calculations accurately most of the time. • Interprets growth, decay, or thresholds reasonably in context. • Uses mathematical evidence to support predictions or conclusions.	• Uses exponential or logarithmic reasoning inconsistently or with limited interpretation. • Thresholds, ratios, or calculations may be unclear or incomplete.	• Little or no meaningful use of exponential or logarithmic reasoning. • Equations, calculations, or interpretations are inaccurate or unrelated to the investigation.
3 POLYNOMIAL BEHAVIOR	• Uses polynomial equations/models with sophisticated understanding to describe complex graph behavior. • Identifies and interprets zeros (including multiplicity), turning points, intervals, concavity, end behavior, and local/global behavior accurately. • Uses equations, graphs, and calculations to explain changing rates, thresholds, and key features in context. • Justifies degree, form, and model choice and critiques limitations.	• Uses polynomial reasoning appropriately. • Identifies and interprets key graph features such as turning points or zeros. • Uses mathematical evidence to explain graph behavior reasonably in context.	• Uses polynomial reasoning with limited interpretation or incomplete calculations. • Connections between graph behavior and context may be weak.	• Little or no meaningful use of polynomial reasoning. • Equations, graph analysis, or interpretations are inaccurate or disconnected from the investigation.
4 MATHEMATICAL INTERPRETATION	• Clearly explains graphs, equations, calculations, and models in context with depth and precision. • Interprets mathematical behavior using accurate and sophisticated vocabulary. • Explains assumptions, limitations, intervals, thresholds, parameter meaning, and uncertainty meaningfully. • Connects mathematics directly to the system and what it reveals or hides.	• Explains mathematical reasoning clearly most of the time. • Connects equations and models reasonably to context. • Uses appropriate mathematical vocabulary.	• Explanations may be incomplete, vague, or inconsistently connected to context. • Interpretation of graphs or equations may be partially developed.	• Mathematical reasoning is missing, unclear, or inaccurate. • Explanations are disconnected from the investigation.
5 REVISION + CRITIQUE	• Revises models, assumptions, predictions, or conclusions meaningfully using strong evidence. • Uses critique, counterexamples, sensitivity analysis, or conflicting evidence thoughtfully. • Explains how understanding changed and why. • Identifies limitations, uncertainty, and confounding variables clearly and evaluates their impact.	• Revises thinking using evidence or feedback appropriately. • Identifies some assumptions or limitations with reasonable explanation.	• Revision is limited or only partially supported by evidence. • Assumptions or limitations may be vague or incomplete.	• Little or no evidence of revision. • Assumptions, limitations, or critique are missing or incorrect.
6 COMMUNICATION + REFLECTION	• Communicates mathematical reasoning clearly, precisely, and thoughtfully in handwritten work and vlog discussions. • Uses multiple representations (equations, graphs, tables, words) effectively. • Reflection demonstrates deep awareness of changing understanding and how mathematics was used. • Contributes consistently and meaningfully to collaborative discussion.	• Communicates reasoning clearly most of the time. • Reflection explains some changes in understanding. • Participates appropriately in discussion and investigation work.	• Communication or reflection may be inconsistent, vague, or underdeveloped. • Participation may be limited.	• Communication, reflection, or participation is minimal or unclear.

GROUP RUBRIC (2 STANDARDS)

STANDARD	4 – EXCEEDS EXPECTATIONS	3 – MEETS EXPECTATIONS	2 – APPROACHING EXPECTATIONS	1 – BELOW EXPECTATIONS
1 INVESTIGATION QUALITY	• Investigation uses appropriate real-world data and mathematically meaningful models. • Equations, graphs, calculations, thresholds, intervals, and variables are interpreted clearly and with sophistication. • Multiple models are compared thoughtfully; model choice is justified. • Predictions and conclusions are supported by strong mathematical evidence. • Confounding variables, assumptions, uncertainty, and limitations are acknowledged clearly and evaluated. • Sources are credible, properly cited, and used responsibly.	• Investigation uses appropriate data and mathematical models. • Graphs, equations, and calculations support conclusions reasonably. • Some limitations and confounding variables are acknowledged. • Sources are credible and used responsibly.	• Investigation uses incomplete or partially appropriate data or models. • Interpretation may be simplistic or weakly supported. • Mathematical reasoning may be inconsistent.	• Investigation lacks meaningful mathematical modeling or evidence-based reasoning. • Conclusions are unsupported or disconnected from the data. • Sources are unreliable or missing.
2 COLLABORATIVE INQUIRY	• Group collaborates consistently throughout the investigation. • Members build on, challenge, and refine ideas using mathematicgl evidence. and revision. • Group uses AI tools responsibly and critically. • All members contribute meaningfully to the final product.	• Group collaborates effectively most of the time. • Members contribute appropriately to discussion, revision, and investigation work. • Shared work shows evidence of collaboration.	• Collaboration or discussion is inconsistent or uneven across members. • Revision or critique may be limited. • Shared work may be incomplete.	• Collaboration is minimal or ineffective. • Little evidence of shared inquiry or meaningful discussion. • Shared work is incomplete.

EXPECTED OUTCOMES: *Use mathematics to construct and interpret models, perform calculations, analyze key features, make predictions, evaluate limitations, and communicatee reasoning about a system under pressure.*